

# The Cognitive Science of Learning and Retention: An Evidence Catalogue for Mathematics Practice in Early Adolescence

The transition from foundational arithmetic to advanced abstract mathematics—which occurs during early adolescence (ages 10–14, corresponding to Polish lower-secondary *klasy* 4–8)—requires learners to shift from executing isolated, automated procedures to managing complex, multi-step problem-solving schemas. The design of instructional software for this demographic must be anchored in empirical cognitive science to effectively mediate this transition. This report synthesizes the empirical evidence concerning how learning, retention, and knowledge transfer occur, bounding the efficacy of these mechanisms based on developmental capacity, prior knowledge, and the intrinsic complexity of the mathematical material. The analysis distinguishes between two operational contexts: the self-directed learner and the parent-supported learner.

## Developmental Caveats (Ages 10–14)

The cognitive and neurobiological profile of a 10- to 14-year-old learner establishes absolute boundary conditions for the application of cognitive science principles. During this developmental window, the prefrontal cortex—the region responsible for executive function, inhibitory control, and the delay of gratification—undergoes significant synaptic pruning and myelination. Consequently, cognitive and self-regulatory capacities that are reliably present in adult populations cannot be assumed in this demographic.

The primary developmental gap lies in metacognitive accuracy and self-regulated learning (SRL)<sup>1</sup>. Early adolescents are highly susceptible to the "fluency illusion"—the mistaken metacognitive belief that because mathematical material is easily processed during passive review (e.g., reading a worked example), it has been permanently committed to long-term memory. Their metacognitive calibration, defined as the correlation between their predicted future performance and their actual future performance, is systematically poor. When left to direct their own learning, 10- to 14-year-olds default to low-effort, low-utility strategies, such as massed practice (cramming) and blocked restudying<sup>3</sup>. They actively avoid high-utility strategies, such as spaced retrieval and interleaved practice, because the latter introduce desirable difficulties that subjectively feel like failure.

Furthermore, delay tolerance remains immature. Spaced repetition relies on the premise that a learner is willing to delay practice to optimize long-term retention, trading the immediate gratification of high short-term performance for long-term durability. A 12-year-old working alone generally lacks the temporal discounting mechanisms required to autonomously sustain this trade-off.

Therefore, a critical distinction must be drawn between the self-directed learner and the parent-supported learner. In the self-directed context, cognitive interventions must be structurally forced by the practice environment, as the learner cannot be relied upon to self-enforce desirable difficulties. In the parent-supported context, the parent effectively acts as a surrogate prefrontal cortex, providing the external executive function necessary to enforce spacing schedules, manage the affective frustration inherent in error-driven learning, and prompt elaborative interrogation. Explicit instruction in self-regulated learning strategies is highly effective for this age group, but it requires active scaffolding<sup>1</sup>.

## Evidence Catalogue

### Retrieval Practice and the Testing Effect

**Principle** — Retrieving information from memory improves long-term retention and transfer significantly more than restudying the same material.

**Evidence strength** — Strong. The testing effect is one of the most robust findings in cognitive psychology, supported by multiple independent meta-analyses, though its efficacy in complex mathematics specifically is subject to important boundary conditions.

**Key studies**

| Author(s) & Year        | Study Focus                                                                  | Source |
|-------------------------|------------------------------------------------------------------------------|--------|
| Adesope et al., 2017    | Comprehensive meta-analysis of the testing effect vs. non-testing conditions | 6      |
| Murray et al., 2025     | Meta-analysis of spaced and retrieval practice specifically in mathematics   | 7      |
| van Gog & Sweller, 2015 | Boundary conditions of the testing effect under high element interactivity   | 8      |
| Bego et al., 2024       | Spaced retrieval practice in STEM and its effect on the achievement gap      | 9      |

**Effect size**

| Outcome Measure                        | Effect Size                      | Source        |
|----------------------------------------|----------------------------------|---------------|
| Overall retention (vs. restudying)     | $g = 0.50$ to $0.61$             | <sup>11</sup> |
| Mathematics learning (vs. restudying)  | $g = 0.18$ (95% CI crosses zero) | <sup>7</sup>  |
| Retention with correct-answer feedback | $g = 0.73$                       | <sup>12</sup> |

**Population tested** — Widely replicated across all ages. Explicitly tested in 10–14-year-olds in STEM and mathematics. However, the effect is smaller and less consistent in applied mathematics compared to vocabulary or declarative fact learning<sup>7</sup>.

**Boundary conditions and failure modes** — The testing effect diminishes or disappears entirely when the material has high "element interactivity" (e.g., complex mathematical problem-solving requiring multiple interacting schemas)<sup>8</sup>. If prior knowledge is too low, the cognitive load of retrieval overwhelms working memory, leading to retrieval failure and zero learning gains. It demands high tolerance for difficulty. If the learner cannot successfully retrieve the information, the test does not act as a learning event unless immediate corrective feedback is provided. The short-term performance during retrieval will be lower than during restudying, which can demotivate a self-directed 12-year-old.

**Practical parameters** — Multiple-choice practice tests yield a larger weighted mean effect size ( $g = 0.70$ ) than short-answer tests ( $g = 0.48$ ) in broad meta-analyses, likely due to reduced cognitive load during the retrieval of complex concepts<sup>6</sup>. A gap of 1 to 6 days between initial learning and the retrieval event yields stronger effects ( $g = 0.82$ ) than gaps of less than one day ( $g = 0.56$ )<sup>6</sup>.

## Spaced Repetition and Distributed Practice

**Principle** — Distributing practice sessions over time produces superior long-term retention compared to massing practice into a single session of equivalent total duration.

**Evidence strength** — Strong. Highly replicated across nearly all domains, including mathematics, with broad and conclusive meta-analytic support.

### Key studies

| Author(s) & Year          | Study Focus                                                      | Source        |
|---------------------------|------------------------------------------------------------------|---------------|
| Cepeda et al., 2006; 2008 | Massive quantitative reviews of distributed practice and optimal | <sup>12</sup> |

|                       |                                                                           |    |
|-----------------------|---------------------------------------------------------------------------|----|
|                       | inter-study intervals                                                     |    |
| Murray et al., 2025   | Meta-analysis of spacing effects in mathematics learning                  | 7  |
| Latimier et al., 2021 | Meta-analytic review of spacing schedules for retrieval practice episodes | 15 |

### Effect size

| Outcome Measure                          | Effect Size          | Source |
|------------------------------------------|----------------------|--------|
| Overall retention (spaced vs. massed)    | $d = 0.46$ to $1.15$ | 12     |
| Mathematics learning (spaced vs. massed) | $g = 0.28$ to $0.43$ | 7      |
| Spaced retrieval vs. massed retrieval    | $g = 0.74$           | 15     |

**Population tested** — Replicated extensively across all age groups, including robust replication in middle-school mathematics<sup>7</sup>.

**Boundary conditions and failure modes** — Spaced practice decreases immediate, short-term performance during the acquisition phase compared to massed practice (cramming), which can demoralize the learner and lower self-efficacy. If the spacing interval is too long relative to the learner's initial memory trace, the information decays completely, rendering the subsequent review session an initial learning session rather than a retrieval event. It requires strict executive function to manage schedules over days or weeks, a capacity often lacking in 12-year-olds without software-enforced pacing or parent-supported structure.

**Practical parameters** — The optimal inter-study interval (ISI) depends directly on the desired retention interval (RI). The literature demonstrates that the optimal study gap is roughly 10% to 20% of the target retention interval<sup>12</sup>. For a test in one week, spacing reviews by 1–2 days is optimal; for retaining information for a year, spacing by 3–5 weeks is optimal<sup>12</sup>.

### Expanding vs. Uniform Spacing Schedules

**Principle** — Expanding spacing schedules, where intervals between reviews gradually increase, produce superior retention compared to uniform spacing schedules, where intervals remain

constant.

**Evidence strength** — Contested (leaning toward Folk regarding its absolute superiority). While historically assumed to be superior in software algorithm design, modern meta-analytic data shows no meaningful advantage over uniform spacing when absolute time is controlled.

### Key studies

| Author(s) & Year               | Study Focus                                                          | Source |
|--------------------------------|----------------------------------------------------------------------|--------|
| Latimier et al., 2021          | Comparative meta-analysis of expanding vs. uniform spacing schedules | 15     |
| Nakata, 2015                   | Effects of expanding and equal spacing on learning                   | 16     |
| Karpicke & Bauernschmidt, 2011 | The role of absolute spacing vs. relative spacing schedules          | 12     |

### Effect size

| Outcome Measure                   | Effect Size                                 | Source |
|-----------------------------------|---------------------------------------------|--------|
| Retention (Expanding vs. Uniform) | $g = 0.034$ (Not statistically significant) | 15     |

**Population tested** — Primarily university students and adults; largely language vocabulary and declarative learning rather than applied mathematics.

**Boundary conditions and failure modes** — Expanding intervals can increase initial retrieval success on the very first attempt (because the first gap is extremely short), but they do not fundamentally alter long-term decay rates better than uniform gaps<sup>15</sup>. For highly complex math problems, expanding schedules may actually induce too much forgetting if the subsequent intervals grow too rapidly, leading to constant reteaching rather than retrieval.

**Practical parameters** — Absolute spacing matters significantly more than relative spacing patterns. Any gap is substantially better than no gap<sup>12</sup>. The specific schedule (1-3-5 days vs. 3-3-3 days) is less important than ensuring the total elapsed time between the first and last practice session is sufficiently long<sup>16</sup>.

## Interleaving vs. Blocked Practice

**Principle** — Alternating different types of mathematical problems during practice (interleaving) improves a learner's ability to transfer knowledge and choose the correct solution strategy

compared to practicing one problem type at a time (blocking).

**Evidence strength** — Strong. Well-replicated specifically within mathematics education and cognitive psychology.

**Key studies**

| Author(s) & Year         | Study Focus                                                                   | Source |
|--------------------------|-------------------------------------------------------------------------------|--------|
| Dunlosky et al., 2013    | Comprehensive review of effective learning techniques, including interleaving | 3      |
| Brunmair & Richter, 2019 | Meta-analysis of interleaved vs. blocked practice                             | 7      |
| Lyle et al., 2020        | Retrieval and spacing/interleaving in engineering and applied mathematics     | 19     |

**Effect size**

| Outcome Measure                                        | Effect Size | Source |
|--------------------------------------------------------|-------------|--------|
| Mathematical problem solving (Interleaved vs. Blocked) | $g = 0.34$  | 7      |

**Population tested** — Middle school and high school students, explicitly heavily tested in mathematics, algebra, and geometry<sup>7</sup>.

**Boundary conditions and failure modes** — Interleaved practice severely depresses performance during the acquisition phase. Learners will make significantly more errors during interleaved practice, which leads to high frustration and a false metacognitive belief that they are not learning (a reversal of the fluency illusion)<sup>3</sup>. Interleaving is entirely ineffective, and potentially harmful, if the learner has not first established a basic, foundational understanding of the individual concepts. It is a strategy for discrimination and transfer, not for initial encoding.

**Practical parameters** — Blocked practice must be used initially upon introducing a novel mathematical concept. Once basic procedural execution is established, practice should immediately transition to interleaved sets containing related but distinct problem types<sup>3</sup>.

## Feedback: Timing, Specificity, and Errors

**Principle** — Providing immediate, correct-answer feedback is universally superior to delayed feedback or exploratory practice without feedback.

**Evidence strength** — Contested. The efficacy of feedback timing and specificity is highly moderated by the learner's prior knowledge level (the aptitude-by-treatment interaction).

### Key studies

| Author(s) & Year                  | Study Focus                                                                                     | Source |
|-----------------------------------|-------------------------------------------------------------------------------------------------|--------|
| Fyfe & Rittle-Johnson, 2012, 2016 | The effects of feedback during exploratory mathematics problem solving; prior knowledge matters | 20     |
| Kulik & Kulik, 1988               | Timing of feedback and its impact on learning                                                   | 23     |

### Effect size

| Outcome Measure                                 | Effect Size                                                | Source |
|-------------------------------------------------|------------------------------------------------------------|--------|
| Low prior knowledge (Feedback vs. No Feedback)  | Positive effect ( <i>not reported</i> as single aggregate) | 20     |
| High prior knowledge (Feedback vs. No Feedback) | Negative effect (Feedback harms performance)               | 20     |

**Population tested** — 2nd to 8th-grade students (encompassing the 10–14 age bracket), specifically tested in mathematics equivalence and problem-solving<sup>23</sup>.

**Boundary conditions and failure modes** — For mathematical novices (low prior knowledge), immediate trial-by-trial feedback is highly beneficial and prevents the encoding of misconceptions and errors. However, for learners with moderate to high prior knowledge, immediate feedback actually *harms* learning and reduces long-term transfer compared to no feedback or delayed feedback<sup>20</sup>. Immediate feedback for advanced learners creates an over-reliance on external validation and prevents them from engaging their own metacognitive error-checking routines.

**Practical parameters** — Feedback systems must dynamically adapt to the learner's mastery

level. Novices require immediate, elaborative feedback. As expertise grows, feedback should be delayed or reduced to corrective/verifier feedback to force the learner to evaluate their own mathematical reasoning.

## Desirable Difficulties and Generation Effects

**Principle** — Self-generating information or answers requires more cognitive effort but results in better retention than passively reading or being provided the exact same information.

**Evidence strength** — Strong. Generating answers is a core tenet of the "desirable difficulties" framework, though bounded by the constraint of the generation task.

### Key studies

| Author(s) & Year      | Study Focus                                                               | Source |
|-----------------------|---------------------------------------------------------------------------|--------|
| Bertsch et al., 2007  | Meta-analytic review of the generation effect                             | 25     |
| McCurdy et al., 2020  | Theories of the generation effect and the impact of generation constraint | 27     |
| Dunlosky et al., 2013 | Elaborative interrogation as a generation mechanism                       | 3      |

### Effect size

| Outcome Measure                        | Effect Size          | Source |
|----------------------------------------|----------------------|--------|
| Overall retention (Generated vs. Read) | $d = 0.40$           | 25     |
| Elaborative Interrogation              | $d = 0.85$ to $2.57$ | 3      |

**Population tested** — Broad demographics, primarily verbal learning and paired associates, but includes arithmetic and problem-solving<sup>29</sup>.

**Boundary conditions and failure modes** — The generation effect diminishes when the constraints on the generation are too tight (e.g., highly constrained single-word fill-in-the-blanks) or when the material's element interactivity is excessively high<sup>27</sup>. If a 12-year-old lacks the prerequisite knowledge to generate an answer, asking them to do so causes cognitive overload and failure to learn.

**Practical parameters** — Best implemented via "elaborative interrogation"—asking the learner



to generate an explanation for *why* a specific mathematical step is true, rather than relying on rote fill-in-the-blank mechanics<sup>3</sup>.

## Cognitive Load Theory: Worked Examples and the Expertise-Reversal Effect

**Principle** — Studying fully worked-out solutions is more effective for novice learners than actively attempting to solve the equivalent problems.

**Evidence strength** — Strong. The worked-example effect is a foundational and exhaustively replicated principle of Cognitive Load Theory (CLT).

### Key studies

| Author(s) & Year        | Study Focus                                                            | Source |
|-------------------------|------------------------------------------------------------------------|--------|
| Sweller, 2010           | Element interactivity and intrinsic cognitive load                     | 31     |
| van Gog & Sweller, 2015 | Interaction between testing effect and high element interactivity      | 8      |
| Chen et al., 2018       | Undesirable difficulty effects in high-element interactivity materials | 30     |

### Effect size

| Outcome Measure                                                   | Effect Size                                                                 | Source |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------|--------|
| Problem solving performance (Worked examples vs. Problem solving) | Large positive effect (often >1.0 standard deviations in early acquisition) | 33     |

**Population tested** — Broadly replicated, particularly dominant in mathematics, physics, and algebraic problem solving for early adolescents<sup>14</sup>.

**Boundary conditions and failure modes** — This principle is strictly bounded by the "Expertise Reversal Effect." Worked examples are highly beneficial for novices because they reduce extraneous cognitive load and prevent working memory from being overwhelmed by means-ends search strategies<sup>32</sup>. However, as the learner gains expertise and automates underlying mathematical schemas, studying worked examples becomes redundant and

actively interferes with those automated schemas, resulting in degraded performance compared to active problem solving.

**Practical parameters** — Instruction should utilize the "completion-problem effect" (a fading strategy). Initial instruction provides a fully worked example. The next iteration provides a problem with only the last step missing. Subsequent iterations remove increasing numbers of steps until the learner is solving the entire problem independently.

## Error-Driven Learning and Productive Failure

**Principle** — Attempting to solve complex, ill-structured mathematical problems prior to receiving explicit instruction leads to deeper conceptual understanding than receiving instruction first.

**Evidence strength** — Strong. Supported by extensive classroom field trials and recent robust meta-analytic reviews.

### Key studies

| Author(s) & Year    | Study Focus                                                                                                        | Source |
|---------------------|--------------------------------------------------------------------------------------------------------------------|--------|
| Sinha & Kapur, 2021 | Meta-analysis of problem solving followed by instruction (PS-I) vs. instruction followed by problem solving (I-PS) | 35     |
| Kapur, 2014, 2016   | Productive failure in mathematical problem solving                                                                 | 36     |

### Effect size

| Outcome Measure                                   | Effect Size                                                               | Source |
|---------------------------------------------------|---------------------------------------------------------------------------|--------|
| Conceptual knowledge and transfer (PS-I vs. I-PS) | $g = 0.36$ to $0.39$ (reaching $0.87$ when adjusted for publication bias) | 35     |

**Population tested** — 8th and 9th-grade mathematics students (13–15 years old), learning concepts like standard deviation, variance, and algebraic modeling<sup>36</sup>.

**Boundary conditions and failure modes** — This principle ("Productive Failure") necessitates that the learner fails to produce the canonical solution. It predictably generates negative affect, frustration, and confusion. If the learner gives up prematurely, or if the environment lacks a facilitator (such as a parent) to manage the emotional regulation required, the failure becomes

*unproductive*. Crucially, it only works if the initial problem-solving phase is followed by explicit, direct instruction that resolves the epistemic gaps generated during the struggle. It does not improve rote procedural fluency better than direct instruction; it solely improves conceptual transfer<sup>36</sup>.

**Practical parameters** — The problem-solving phase should last long enough to exhaust the learner's intuitive strategies and make them aware of their knowledge gaps, but must be followed immediately by targeted instruction that organizes the chaotic schemas they generated.

## Metacognition, Self-Regulated Learning, and Calibration

**Principle** — Explicit instruction in self-regulated learning (SRL) strategies improves academic achievement across subject matters.

**Evidence strength** — Strong. Documented via extensive meta-analyses focused specifically on primary and secondary education.

### Key studies

| Author(s) & Year        | Study Focus                                                                                    | Source       |
|-------------------------|------------------------------------------------------------------------------------------------|--------------|
| Dignath & Büttner, 2008 | Components of fostering self-regulated learning among students at primary and secondary levels | <sup>1</sup> |
| Theobald, 2021          | Meta-analysis of SRL training programs                                                         | <sup>2</sup> |

### Effect size

| Outcome Measure                                 | Effect Size           | Source        |
|-------------------------------------------------|-----------------------|---------------|
| Academic achievement (Overall SRL intervention) | $g = 0.69$ to $0.728$ | <sup>1</sup>  |
| Academic achievement (Mathematics specifically) | $d = 0.859$           | <sup>42</sup> |

**Population tested** — Primary and secondary school students (highly relevant to the 10–14 age bracket); heavily tested in mathematics<sup>1</sup>.

**Boundary conditions and failure modes** — SRL cannot be passively absorbed; it must be explicitly modeled. Students will not naturally adopt planning, monitoring, and evaluation strategies. The intervention is heavily dependent on the fidelity of the

instruction—researcher-led interventions historically yield higher effects than teacher-led ones, indicating that poor implementation degrades the effect<sup>1</sup>. Interventions focusing on single strategies (e.g., only cognitive elaboration) are less effective than multi-component interventions.

**Practical parameters** — Interventions must actively combine cognitive strategies (e.g., elaboration), metacognitive strategies (e.g., self-testing, planning), and motivational strategies (e.g., self-efficacy reinforcement) to be fully effective<sup>1</sup>.

## Mastery Learning, Criterion Thresholds, and Overlearning

**Principle** — Requiring learners to demonstrate a high standard of proficiency on a topic before advancing to the next topic yields superior long-term achievement compared to fixed-time pacing.

**Evidence strength** — Moderate. The effect is well-documented, but operational implementations vary wildly, and strict interpretation of Bloom's original estimates are contested as overly optimistic for scalable software.

### Key studies

| Author(s) & Year      | Study Focus                                   | Source |
|-----------------------|-----------------------------------------------|--------|
| Guskey & Pigott, 1988 | Meta-analysis of group-based mastery learning | 43     |
| Kulik et al., 1990    | Effectiveness of mastery learning programs    | 43     |
| Bloom, 1984           | The 2-Sigma problem                           | 43     |

### Effect size

| Outcome Measure                                 | Effect Size                            | Source |
|-------------------------------------------------|----------------------------------------|--------|
| Academic achievement (Mastery vs. Conventional) | $d = 0.52$ to $0.59$                   | 43     |
| Tutoring (Bloom's theoretical bound)            | $d = 2.0$ (rarely replicated at scale) | 45     |

**Population tested** — K-12 education, highly effective in sequentially structured subjects like mathematics where knowledge strictly stacks<sup>43</sup>.

**Boundary conditions and failure modes** — If the mastery threshold is set too low (e.g., 60%), it permits the accumulation of knowledge gaps that cripple future learning in cumulative subjects like math<sup>45</sup>. If the "corrective instruction" provided after a failed mastery check is simply a repetition of the exact same instruction at a slower pace, it fails. Furthermore, insisting on 100% immediate mastery often leads to "overlearning," which provides rapidly diminishing returns for long-term retention compared to spacing<sup>47</sup>. Time costs can be severe; slower learners may require significantly more time to reach mastery, pulling against standardized curricular schedules.

**Practical parameters** — The literature supports an 80% to 90% accuracy threshold on formative, performance-based assessments to constitute "mastery" before advancing<sup>43</sup>. Corrective instruction must utilize a different pedagogical approach than the initial instruction.

## Consolidation and Wakeful Rest

**Principle** — A brief period of quiet, wakeful rest immediately following a learning phase enhances memory consolidation and reduces forgetting.

**Evidence strength** — Moderate. The core biological mechanism is recognized, but ecological validity in real-world, noisy environments is highly debated.

### Key studies

| Author(s) & Year         | Study Focus                                               | Source |
|--------------------------|-----------------------------------------------------------|--------|
| Dewar et al., 2012, 2014 | Wakeful rest promotes the integration of spatial memories | 48     |
| Craig et al., 2016       | Post-learning rest and cognitive mapping                  | 50     |

### Effect size

| Outcome Measure  | Effect Size                                                       | Source |
|------------------|-------------------------------------------------------------------|--------|
| Memory retention | Highly variable ( <i>not reported</i> as meta-analytic aggregate) | 51     |

**Population tested** — Predominantly adults in controlled laboratory environments, focusing on spatial memory, word lists, and language<sup>49</sup>.

**Boundary conditions and failure modes** — The effect is extremely fragile. It is instantly nullified if the learner engages in cognitive tasks, uses a smartphone, or consumes media during the rest period<sup>51</sup>. For a 12-year-old, sitting in silence without external stimuli is highly

unnatural and exceptionally difficult to enforce in a self-directed environment.

**Practical parameters** — Requires roughly 10 minutes of wakeful rest (eyes closed, no devices, no reading) immediately following intense cognitive encoding<sup>54</sup>.

## Mathematics Anxiety and Affective Load (*Added Area*)

*Rationale for inclusion: Affective states directly mediate cognitive load and retention in mathematics specifically for early adolescents.*

**Principle** — Mathematics anxiety consumes limited working memory capacity, degrading computational performance and learning independently of the learner's actual mathematical aptitude.

**Evidence strength** — Strong. Supported by recent comprehensive meta-analyses.

### Key studies

| Author(s) & Year     | Study Focus                                                             | Source |
|----------------------|-------------------------------------------------------------------------|--------|
| Barroso et al., 2021 | Meta-analysis of the relation between math anxiety and math achievement | 55     |

### Effect size

| Outcome Measure                   | Effect Size                                  | Source |
|-----------------------------------|----------------------------------------------|--------|
| Math performance and Math Anxiety | $g = -0.467$ (moderate negative correlation) | 55     |

**Population tested** — School-aged students (childhood and adolescence)<sup>57</sup>.

**Boundary conditions and failure modes** — Math anxiety creates extraneous cognitive load. If a learner is highly anxious, interventions that increase difficulty (like interleaving or productive failure) may trigger a panic response, completely halting learning<sup>56</sup>.

**Practical parameters** — Cognitive interventions that utilize desirable difficulties must be paired with affective regulation (e.g., explicit reframing of failure as a normative learning tool) to free up working memory resources prior to complex problem solving.

## Adjudication of Widely Circulated Claims

### "Learning Styles" (Visual, Auditory, Kinesthetic)

- **Rating:** Folk (Myth).
- **Adjudication:** The "meshing hypothesis"—the idea that instruction is most effective when it matches a learner's self-reported preferred modality (e.g., presenting pictures to a "visual learner")—has absolutely no empirical support<sup>59</sup>. Rigorous experimental designs

consistently fail to find the required crossover interaction<sup>61</sup>. Attempting to restrict a 12-year-old's math instruction to an "auditory" style is not only ineffective but potentially harmful by limiting their exposure to the optimal modality for the specific subject matter (e.g., geometry requires visual/spatial representation, regardless of learner preference)<sup>59</sup>.

### **The Forgetting Curve (as commonly drawn)**

- **Rating:** Folk (in its rigid, mathematical application to complex subjects).
- **Adjudication:** While Ebbinghaus's observation of exponential memory decay is a foundational concept, the popular graphs showing knowledge dropping to 20% in three days are based on the memorization of meaningless nonsense syllables. Meaningful, highly structured information (like algebraic schemas) decays much more slowly and is heavily moderated by prior knowledge and element interactivity. Treating the forgetting curve as an immutable biological clock for scheduling math software reviews ignores the semantic weight of the material.

### **"Neuroplasticity" as a basis for learning advice**

- **Rating:** Folk (as a prescriptive tool).
- **Adjudication:** While neuroplasticity is the biological mechanism of all learning, it is a descriptive biological reality, not a prescriptive psychological intervention. Using "neuroplasticity" to justify specific study habits (e.g., "brain training" games) is a neuro-mythological leap. Cognitive architectures (like Cognitive Load Theory) provide actual predictive models for instruction; neuroplasticity merely confirms that the brain changes when learning occurs.

### **1% Better / Compounding Gains Framing**

- **Rating:** Folk / Contested.
- **Adjudication:** The popular self-help narrative that learning compounds exponentially (1% better every day) directly contradicts the cognitive science of skill acquisition. Human learning follows a "Power Law of Practice"<sup>63</sup>. Performance improves rapidly at first, and then the rate of improvement slows down dramatically, resulting in a negatively accelerated curve (diminishing returns)<sup>63</sup>. A learner practicing math will experience a plateau, requiring immense effort for marginal gains later on, not exponential compounding<sup>66</sup>.

### **Working Memory Training / Dual N-Back**

- **Rating:** Weak.
- **Adjudication:** Extensive literature shows that practicing working memory games (like dual n-back) makes users better at those specific games (near transfer), but it does not expand fundamental working memory capacity, nor does it transfer to complex academic skills like mathematics (far transfer). Working memory limits are biological and largely fixed; instruction must adapt to the limits (via Cognitive Load Theory), as the limits cannot be expanded through training.

## **Conflicts and Open Questions**

Translating these cognitive principles into structural software decisions reveals several inherent theoretical tensions. A practitioner must balance these conflicting forces rather than viewing

the catalogue as a checklist of universally compatible rules.

**Immediate Feedback vs. Desirable Difficulty** There is a direct structural conflict between the necessity of correcting errors to prevent the encoding of misconceptions and the necessity of making learning difficult to improve retention. Providing immediate, correct-answer feedback reduces cognitive load and ensures accuracy, which is vital for mathematical novices<sup>20</sup>.

However, immediate feedback acts as a cognitive crutch; it bypasses the retrieval effort and prevents the learner from engaging their own metacognitive monitoring. The literature resolves this via the expertise-reversal effect: feedback must be faded and delayed as the learner's proficiency increases, intentionally introducing difficulty back into the system to force independent error-checking<sup>22</sup>.

**Reducing Cognitive Load vs. Productive Failure** Cognitive Load Theory (CLT) argues that a novice's working memory must be protected from extraneous load, advocating for fully worked examples and explicit, direct instruction<sup>32</sup>. Conversely, Productive Failure deliberately plunges learners into complex, ill-structured problems without prior instruction, overwhelming working memory to induce failure<sup>38</sup>. These paradigms pull in opposite directions. The emerging synthesis suggests a sequential resolution: Productive failure is highly effective for *schema activation and conceptual understanding* when introducing a radically new concept (e.g., moving from arithmetic to statistical variance) because it highlights epistemic gaps<sup>36</sup>. However, once the explicit instruction is subsequently delivered, CLT must govern the *procedural acquisition* phase, utilizing worked examples to reduce load while the learner masters the mechanics<sup>68</sup>.

**Spacing vs. Mastery Criteria** Mastery learning requires a learner to reach a high criterion (e.g., 90%) before moving on, heavily focusing on massed or tightly looped corrective instruction<sup>43</sup>. Spacing requires abandoning a topic before it is perfectly mastered to allow time to pass before revisiting it<sup>12</sup>. Insisting on 100% immediate mastery often leads to overlearning, which provides rapidly diminishing returns for long-term retention<sup>47</sup>. The tension is resolved by aiming for a "sufficient" initial mastery threshold (e.g., 80%), advancing to a new topic (which introduces interleaving), and relying on spaced retrieval to close the remaining gap and cement the knowledge over time<sup>43</sup>.

**High Element Interactivity vs. The Testing Effect** Retrieval practice (the testing effect) is frequently celebrated as a universal good, but van Gog and Sweller (2015) warn that when mathematical concepts have high "element interactivity" (requiring the simultaneous processing of many dependent variables in working memory), retrieval practice can actually yield worse results than restudying<sup>8</sup>. If a 12-year-old cannot hold the required mathematical elements in working memory, a test does not trigger neural reconsolidation; it triggers cognitive overload<sup>31</sup>. In these cases, restudying a worked example produces better long-term retention than attempting and failing to retrieve a complex schema<sup>30</sup>. Practitioners must deconstruct complex math problems into isolated, low-interactivity elements for initial retrieval practice, only testing the fully integrated schema once the sub-components are automated.

## Works cited



1. (PDF) Components of fostering self-regulated learning among, [https://www.researchgate.net/publication/225530007\\_Components\\_of\\_fostering\\_self-regulated\\_learning\\_among\\_students\\_A\\_meta-analysis\\_on\\_intervention\\_studies\\_at\\_primary\\_and\\_secondary\\_school\\_level](https://www.researchgate.net/publication/225530007_Components_of_fostering_self-regulated_learning_among_students_A_meta-analysis_on_intervention_studies_at_primary_and_secondary_school_level)
2. The effectiveness of self-regulated learning (SRL) interventions on, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9650592/>
3. Improving Students' Learning with Effective Learning Techniques, <https://dev.taleafrica.com/2021/03/29/improving-students-learning-with-effective-learning-techniques-promising-directions-from-cognitive-and-educational-psychology/>
4. Improving Students' Learning with Effective Study Techniques – Part 1, <https://uminntilt.com/2014/03/03/improving-students-learning-1/>
5. Full article: A meta-analysis of the efficacy of self-regulated learning, <https://www.tandfonline.com/doi/full/10.1080/0144929X.2022.2151935>
6. Important new meta-analysis on the testing effect, <https://theeconomyofmeaning.com/2017/03/21/important-new-meta-analysis-on-the-testing-effect-with-some-surprises/>
7. A Meta-analytic Review of the Effectiveness of Spacing and, [http://aidanhorner.org/papers/Murrayetal\\_EdPsychReview\\_2025.pdf](http://aidanhorner.org/papers/Murrayetal_EdPsychReview_2025.pdf)
8. (PDF) Not New, but Nearly Forgotten: the Testing Effect Decreases, [https://www.researchgate.net/publication/277908827\\_Not\\_New\\_but\\_Nearly\\_Forgotten\\_the\\_Testing\\_Effect\\_Decreases\\_or\\_even\\_Disappears\\_as\\_the\\_Complexity\\_of\\_Learning\\_Materials\\_Increases](https://www.researchgate.net/publication/277908827_Not_New_but_Nearly_Forgotten_the_Testing_Effect_Decreases_or_even_Disappears_as_the_Complexity_of_Learning_Materials_Increases)
9. Retrieval practice – a tool to be able to retain higher mathematics, [https://www.researchgate.net/publication/380553709\\_Retrieval\\_practice\\_-\\_a\\_tool\\_to\\_be\\_able\\_to\\_retain\\_higher\\_mathematics\\_even\\_3\\_months\\_after\\_the\\_exam](https://www.researchgate.net/publication/380553709_Retrieval_practice_-_a_tool_to_be_able_to_retain_higher_mathematics_even_3_months_after_the_exam)
10. (PDF) Single-paper meta-analyses of the effects of spaced retrieval, [https://www.researchgate.net/publication/378038633\\_Single-paper\\_meta-analyses\\_of\\_the\\_effects\\_of\\_spaced\\_retrieval\\_practice\\_in\\_nine\\_introduutory\\_STEM\\_courses\\_is\\_the\\_glass\\_half\\_full\\_or\\_half\\_empty](https://www.researchgate.net/publication/378038633_Single-paper_meta-analyses_of_the_effects_of_spaced_retrieval_practice_in_nine_introduutory_STEM_courses_is_the_glass_half_full_or_half_empty)
11. Testing the testing effect on prolific: when retrieval practice fails to, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12894256/>
12. Spaced Repetition: 130+ Years of Research, <https://memormore.app/blogs/spaced-repetition-research>
13. van Gog, T., & Sweller, J. (2015). Not New, but Nearly Forgotten, <https://notes.andymatuschak.org/zMjGz9ddgwjxadBRxQCnnWY>
14. (PDF) High Element Interactivity Information During Problem Solving, [https://www.researchgate.net/publication/273297410\\_High\\_Element\\_Interactivity\\_Information\\_During\\_Problem\\_Solving\\_may\\_Lead\\_to\\_Failure\\_to\\_Obtain\\_the\\_Testing\\_Effect](https://www.researchgate.net/publication/273297410_High_Element_Interactivity_Information_During_Problem_Solving_may_Lead_to_Failure_to_Obtain_the_Testing_Effect)
15. A Meta-Analytic Review of the Benefit of Spacing out Retrieval, [https://www.researchgate.net/publication/344611197\\_A\\_Meta-Analytic\\_Review\\_of\\_the\\_Benefit\\_of\\_Spacing\\_out\\_Retrieval\\_Practice\\_Episodes\\_on\\_Retention](https://www.researchgate.net/publication/344611197_A_Meta-Analytic_Review_of_the_Benefit_of_Spacing_out_Retrieval_Practice_Episodes_on_Retention)
16. Effects of expanding and equal spacing on second language, [https://www.researchgate.net/publication/277612362\\_EFFECTS\\_OF\\_EXPANDING](https://www.researchgate.net/publication/277612362_EFFECTS_OF_EXPANDING)

AND\_EQUAL\_SPACING\_ON\_SECOND\_LANGUAGE\_VOCABULARY\_LEARNING\_Do  
es\_Gradually\_Increasing\_Spacing\_Increase\_Vocabulary\_Learning

17. Spaced repetition - Wikipedia, [https://en.wikipedia.org/wiki/Spaced\\_repetition](https://en.wikipedia.org/wiki/Spaced_repetition)
18. Optimising retention through multiple study opportunities over days, [https://www.researchgate.net/publication/264796869\\_Optimising\\_retention\\_through\\_multiple\\_study\\_opportunities\\_over\\_days\\_The\\_benefit\\_of\\_an\\_expanding\\_schedule\\_of\\_repetitions](https://www.researchgate.net/publication/264796869_Optimising_retention_through_multiple_study_opportunities_over_days_The_benefit_of_an_expanding_schedule_of_repetitions)
19. Retrieval practice and spacing in an engineering mathematics, [https://www.researchgate.net/publication/321817813\\_Retrieval\\_practice\\_and\\_spacing\\_in\\_an\\_engineering\\_mathematics\\_classroom\\_Do\\_the\\_effects\\_add\\_up](https://www.researchgate.net/publication/321817813_Retrieval_practice_and_spacing_in_an_engineering_mathematics_classroom_Do_the_effects_add_up)
20. The Effects of Feedback During Exploration Depend on Prior, [https://cdn.vanderbilt.edu/vu-sub/wp-content/uploads/sites/280/2023/08/04182512/ATME\\_Fyfe\\_CogSciPaper\\_2012\\_Final.pdf](https://cdn.vanderbilt.edu/vu-sub/wp-content/uploads/sites/280/2023/08/04182512/ATME_Fyfe_CogSciPaper_2012_Final.pdf)
21. The benefits of computer-generated feedback for mathematics, <https://pubmed.ncbi.nlm.nih.gov/27082020/>
22. The benefits of computer-generated feedback for mathematics, <https://files.eric.ed.gov/fulltext/ED566264.pdf>
23. The Effect of Feedback Timing on Mathematics Achievement Geri, <https://ilkogretim-online.org/index.php/pub/article/download/1397/1357/2684>
24. The Timing of Feedback on Mathematics Problem Solving in ... - ERIC, <https://eric.ed.gov/?id=ED562489>
25. (PDF) The generation effect: A meta-analytic review - ResearchGate, [https://www.researchgate.net/publication/6193801\\_The\\_generation\\_effect\\_A\\_meta-analytic\\_review](https://www.researchgate.net/publication/6193801_The_generation_effect_A_meta-analytic_review)
26. The generation effect: a meta-analytic review - PubMed, <https://pubmed.ncbi.nlm.nih.gov/17645161/>
27. Theories of the Generation Effect and the Impact of ... - ResearchGate, [https://www.researchgate.net/publication/346204818\\_Theories\\_of\\_the\\_Generation\\_Effect\\_and\\_the\\_Impact\\_of\\_Generation\\_Constraint\\_A\\_Meta-analytic\\_Review](https://www.researchgate.net/publication/346204818_Theories_of_the_Generation_Effect_and_the_Impact_of_Generation_Constraint_A_Meta-analytic_Review)
28. Theories of the generation effect and the impact of ... - PubMed, <https://pubmed.ncbi.nlm.nih.gov/32671573/>
29. The Generation Effect: Activating Broad Neural Circuits During, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3556209/>
30. Element interactivity - Andy Matuschak's notes, <https://notes.andymatuschak.org/zBk3AaWYp4spoTbc1f5aYvC>
31. Not New, but Nearly Forgotten - Utrecht University Repository, <https://dspace.library.uu.nl/server/api/core/bitstreams/390bc589-b82a-4557-9484-8e84c8331e20/content>
32. Chapter 02: Cognitive Load Theory and Instructional Message Design, [https://digitalcommons.odu.edu/context/instructional\\_message\\_design\\_vol2/article/1001/viewcontent/2\\_InstructionalMessageDesign\\_vol2\\_Shaffer.pdf](https://digitalcommons.odu.edu/context/instructional_message_design_vol2/article/1001/viewcontent/2_InstructionalMessageDesign_vol2_Shaffer.pdf)
33. The Worked-Example Effect and a Mastery Approach Goal Orientation, [https://www.researchgate.net/publication/381141116\\_The\\_Worked-Example\\_Effect\\_and\\_a\\_Mastery\\_Approach\\_Goal\\_Orientation](https://www.researchgate.net/publication/381141116_The_Worked-Example_Effect_and_a_Mastery_Approach_Goal_Orientation)
34. Worked-example effect - Wikipedia,

- [https://en.wikipedia.org/wiki/Worked-example\\_effect](https://en.wikipedia.org/wiki/Worked-example_effect)
35. Productive failure (errors) Details - Visible Learning MetaX,  
[https://www.visiblelearningmetax.com/influences/view/productive\\_failure\\_\(errors\)](https://www.visiblelearningmetax.com/influences/view/productive_failure_(errors))
  36. Every Classroom Teaches the Lesson Before Assigning Problems. A,  
<https://www.studiessuggest.net/stories/failing-first-improves-learning.html>
  37. When Problem Solving Followed by Instruction Works: Evidence for,  
<https://www.semanticscholar.org/paper/When-Problem-Solving-Followed-by-Instruction-Works%3A-Sinha-Kapur/4712a08981b614f2a06399d4da4bab7636643d78>
  38. Productive failure in learning math - PubMed,  
<https://pubmed.ncbi.nlm.nih.gov/24628487/>
  39. Enriching problem-solving followed by instruction with explanatory,  
<https://www.tandfonline.com/doi/full/10.1080/10508406.2021.1964506>
  40. International Online Journal of Educational Sciences,  
[https://www.iojes.net/?mod=makale\\_tr\\_ozet&makale\\_id=49642](https://www.iojes.net/?mod=makale_tr_ozet&makale_id=49642)
  41. Impact of Self-Regulated Learning Training | PDF | Meta Analysis,  
<https://www.scribd.com/document/806717389/Self-regulated-learning-training-programs-enhance-university-students-academic-performance-self-regulated-learning-strategies-and-motivation-A-met>
  42. (PDF) The Effect of Self-Regulated Learning Strategies on Academic,  
[https://www.researchgate.net/publication/318193952\\_The\\_Effect\\_of\\_Self-Regulated\\_Learning\\_Strategies\\_on\\_Academic\\_Achievement\\_A\\_Meta-Analysis\\_Study](https://www.researchgate.net/publication/318193952_The_Effect_of_Self-Regulated_Learning_Strategies_on_Academic_Achievement_A_Meta-Analysis_Study)
  43. Mastery Learning: Definition, Examples and Evidence (2026),  
<https://www.structural-learning.com/post/mastery-learning>
  44. EJ378253 - Research on Group-Based Mastery Learning Programs,  
<https://eric.ed.gov/?id=EJ378253>
  45. Mastery learning and Bloom's two-sigma problem - Wiz Kids,  
<https://wiz.kids/learn/science-of-learning/mastery-learning>
  46. A Practical Review of Mastery Learning - PMC,  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10159400/>
  47. Long-Term Retention Explained by a Model of Short-Term Learning,  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2585394/>
  48. The effects of wakeful rest on memory consolidation in an online,  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9642069/>
  49. Wakeful rest promotes the integration of spatial memories into,  
[https://www.researchgate.net/publication/280630164\\_Wakeful\\_Rest\\_Promotes\\_the\\_Integration\\_of\\_Spatial\\_Memories\\_Into\\_Accurate\\_Cognitive\\_Maps](https://www.researchgate.net/publication/280630164_Wakeful_Rest_Promotes_the_Integration_of_Spatial_Memories_Into_Accurate_Cognitive_Maps)
  50. Designing for Memory: How Pathway ... - Research Square,  
<https://assets-eu.researchsquare.com/files/rs-8882289/v1/111b3b6d-801f-4ba0-a773-a3fcad534a14.pdf?c=1772443782>
  51. Wakeful rest and memory consolidation in an ecologically valid,  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12945945/>
  52. Wakeful rest promotes the integration of spatial memories into,  
<https://scispace.com/pdf/wakeful-rest-promotes-the-integration-of-spatial-memories-4u1fv5qd77.pdf>
  53. Resting after learning or repeating the learned?,

<https://wlv.openrepository.com/server/api/core/bitstreams/6c863ade-a39e-4235-9f33-cb7de4f5f9c7/content>

54. Wakeful Rest Benefits Recall, but Not Recognition, of Incidentally,  
<https://www.mdpi.com/2076-3425/12/12/1609>
55. A Meta-Analysis of Math Anxiety Interventions,  
<https://jnc.psychopen.eu/index.php/jnc/article/view/8401/8401.html>
56. Math Anxiety, Math Performance and Role of Field Experience in,  
<https://www.mdpi.com/2227-7102/15/9/1227>
57. The relation between mathematics anxiety and ... - ResearchGate,  
[https://www.researchgate.net/publication/337062596\\_The\\_relation\\_between\\_mathematics\\_anxiety\\_and\\_mathematics\\_performance\\_among\\_school-aged\\_students\\_A\\_meta-analysis](https://www.researchgate.net/publication/337062596_The_relation_between_mathematics_anxiety_and_mathematics_performance_among_school-aged_students_A_meta-analysis)
58. Disentangling the individual and contextual effects of math anxiety,  
<https://www.pnas.org/doi/10.1073/pnas.2115855119>
59. Learning Styles: Out of Fashion - American Psychological Association,  
<https://www.apa.org/pubs/highlights/spotlight/issue-22>
60. (PDF) Learning Styles: Concepts and Evidence - ResearchGate,  
[https://www.researchgate.net/publication/233600402\\_Learning\\_Styles\\_Concepts\\_and\\_Evidence](https://www.researchgate.net/publication/233600402_Learning_Styles_Concepts_and_Evidence)
61. The Modality-Specific Learning Style Hypothesis: A Mini-Review,  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC6113575/>
62. Learning Styles, Preferences, or Strategies? An Explanation for the,  
<https://www.teachertoolkit.co.uk/wp-content/uploads/2025/04/s10648-025-10002-w.pdf>
63. The learning curve - Frank Ritter, <https://www.frankritter.com/papers/ritterS01.pdf>
64. Piecewise power laws in individual learning curves - PMC - NIH,  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4577530/>
65. Rethinking Grading Systems Through the Lens of the Power Law of,  
<https://www.geyerinstructional.com/blog/rethinking-grading-systems-through-the-lens-of-the-power-law-of-learning/>
66. Mathematical model bridges disparate timescales of lifelong learning,  
<https://arxiv.org/html/2206.03954v1>
67. (PDF) Productive Failure - ResearchGate,  
[https://www.researchgate.net/publication/381913649\\_Productive\\_Failure](https://www.researchgate.net/publication/381913649_Productive_Failure)
68. From problem-solving to sensemaking: A comparative meta-analysis,  
[https://www.researchgate.net/publication/351889202\\_From\\_problem-solving\\_to\\_sensemaking\\_A\\_comparative\\_meta-analysis\\_of\\_preparatory\\_approaches\\_for\\_future\\_learning](https://www.researchgate.net/publication/351889202_From_problem-solving_to_sensemaking_A_comparative_meta-analysis_of_preparatory_approaches_for_future_learning)
69. How the Amount and Spacing of Retrieval Practice Affect the Short,  
<https://www.semanticscholar.org/paper/How-the-Amount-and-Spacing-of-Retrieval-Practice-of-Lyle-Bego/e28f53b76e36fc07f7fecc5f5233497a2f985d14>